SPECTRA SOLAR PARK LTD _35 MW AC GRID-TIED BY SPECTRA SOLAR PARK LIMITED

EKI Energy Services Limited

Project Title	*Spectra Solar Park Ltd _35 MW AC Grid-Tied by Spectra Solar Park Limited*
Version	*01*
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Prepared By	*EKI Energy Services Ltd.*
Contact	*Address: Office no. 201, Plot 48, Scheme 78 part 2 Vijay Nagar, Near Brilliant Convention Centre Indore - 452010 (M.P, India)* *Email ID:* registry@enkingint.org *Website* *www.enkingint.org*

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Implemented with the objective of generating clean energy (electricity) by harnessing renewable energy (solar) resources and curbing emission of greenhouse gas, the proposed project activity by Spectra Solar Park Limited includes deployment of grid connected Solar Power projects with cumulative capacity of 35 MW (commissioned on 12th March 2021).

A summary description of the technologies/ measures to be implemented by the project

The 35 MW grid tied solar power project comprising of Solar PV module (Poly-crystalline), grid tied inverter, power transformer, power evacuation infrastructure, energy metering points, switch yards and other civil constructions is implemented at a site where no other power plant was operating prior to the implementation of the project activity (green-field power plant). Electricity generated from the project activity is supplied to Bangladesh Power Development Board, thus substitutes high carbon intensive grid electricity by clean and renewable based electricity.

Location of the project

The 35 MW solar Power Project is located at Shibaloy, Upazila of Manikgang District under Dhaka Division of Bangladesh. The detailed location is provided in section 1.12 of this document.

Explanation on how the project is expected to generate GHG emission reductions

The electricity supplied to the Bangladesh grid from the operational solar power plant, thus replaces/displaces the feeding of equivalent amount of electricity generated from the operation of existing grid connected power plants (mostly fossil fuel based) and anthropogenic emissions of greenhouse gases (GHGs) associated with its generation. Since the project activity is a green field power generation initiative, therefore in accordance to the applied methodology the baseline to the project activity is "electricity delivered to the grid by the operation of grid-connected power plants and by the addition of new generation sources",

Scenario existing prior to the implementation of the project

Bangladesh power sector is highly reliant on fossil fuel-based power plant using natural gas (51.97%), furnace fuel (27.25%), diesel (5.86%) and coal (8.03%). As on 2020-21[1] out of the total installed capacity of 22,031 MW, the share of renewable includes 230 MW (1.04%) of Hydro power project and 129 MW (0.59%) solar power project. Therefore, the situation prior to the implementation of the project activity is generation of electricity in the fossil fuel fired power plant and its supply to national grid.

Estimate of annual average and total GHG emission reductions and removals

The project is expected to result in an average annual supply of 60,280 MWh/ year[2] and cumulative supply of 421,962 MWh of green power to the Bangladesh grid during the first crediting period. The average annual emission reduction from the project is estimated to be

[1] Bangladesh Power Development Board, Annual Report 2020-21

[2] Average of seven years

around 39,694 tCO_{2e}/annum and a cumulative emission reduction of 277,857 tCO_{2e} for the first crediting period.

1.2 Sectoral Scope and Project Type

The sectoral scope and type of project applicable are as below

Sectoral scope: 01 - Energy industries (renewable / non-renewable sources)

Type: I – Renewable Energy Projects

Methodology - ACM0002 Grid-connected electricity generation from renewable sources --- Version 20.0[3]

The project activity is not a grouped project

1.3 Project Eligibility

The project activity involves generation of electricity using solar energy resources that result in avoidance of fossil fuel-based power generation and associated emission of carbon-di-oxide (one of the Six Kyoto Protocol greenhouse gases).

Additionally, the project falls under Sectoral Scope -I Energy industries (renewable / non-renewable sources) and is supported by methodology approved under an approved under CDM program.

As per VCS Standard Version 4.2, section 2.1.1, since the project activity involves Grid-connected electricity generation using solar power plants and is located in Bangladesh (LDC country[4]) the project activity is eligible under VCS project.

Therefore, the project is eligible under scope of VCS program.

1.4 Project Design

The project is not a grouped project and designed to include a single installation of an activity.

Eligibility Criteria

Not Applicable

1.5 Project Proponent

Organization name	Spectra Solar Park Limited
Contact person	Khan Md. Aftabuddin
Title	Managing Director

[3] https://cdm.unfccc.int/methodologies/DB/XP2LKUSA61DKUQC0PIWPGWDN8ED5PG

[4] UN list of least developed countries | UNCTAD

Address	House: 17, Road: 106, Block: CEN(F), Gulshan-2, Dhaka-1212, Bangladesh
Telephone	880-255059854, 880-2-58816192
Email	chairmanspectragroup@gmail.com

1.6 Other Entities Involved in the Project

Organization name	EKI Energy Services Limited
Role in the project	Project Consultant
Contact person	Manish Dabkara
Title	Managing Director & Chief Executive Officer
Address	Office No 201, Plot No 48, Scheme 78, Vijay Nagar Part II, Indore 452010, India
Telephone	+91 99075 34900
Email	registry@enkingint.org

1.7 Ownership

The project ownership is with Spectra Solar Park Limited. The ownership of the project proponent, i.e Spectra Solar Park Limited is demonstrated through the commissioning certificate and Power Purchase Agreement signed with BPDB (Bangladesh Power Development Board).

1.8 Project Start Date

Project Start Date: 12/03/2021

The project start date is the date on which project activity was commissioned (synchronisation with the grid and start of GHG emission reductions).

1.9 Project Crediting Period

Crediting Period Start date:12/03/2021

Crediting Period End date: 11/03/2028

Estimated life time of the project Instance 1 technology equipment:

25 Years 00 Months

The project activity adopts renewable crediting period of 7 years period which can be renewed for maximum 2 times.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Since the emission reduction under the project activity is less than 300,000 tCO2e/year the project activity instance is classified as "Projects".

Project Scale	
Project	✓
Large project	

Year	Estimated GHG emission reductions or removals (tCO_2e)
Year 1[5]	40,535
Year 2	40,252
Year 3	39,970
Year 4	39,690
Year 5	39,412
Year 6	39,136
Year 7	38,862
Total estimated ERs	277,857
Total number of crediting years	7
Average annual ERs	39,694

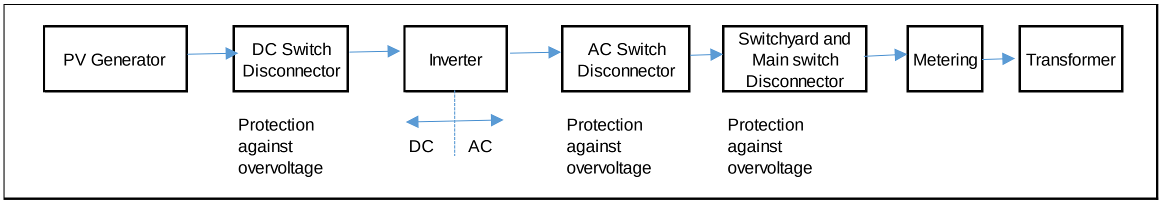
1.11 Description of the Project Activity

The project activity involves installation of grid connected Solar PV based electricity generation units with a cumulative capacity of 35 MW in Bangladesh. The solar PV-grid connected system produces DC power in Solar PV array (PV modules are connected in series and parallel to form an array to get the desired DC voltage and power. Modules in series, called strings, are grouped together through appropriate combiner boxes (AJB/SMU) to ensure that the inverter inputs match with the output of the combiner boxes), converted to AC supply through inverters and step it up to an appropriate level for transmission and evacuation at the designated utility substation. The inverter/power conditioning unit extracts the maximum available power from the solar array and feeds it to the grid. If the grid voltage and/or frequency is out of the range, the PCU immediately isolates the system from the grid. Metering is carried out at solar power plant level through main

[5] Year one starts from the date of commissioning of first instance of this grouped project activity i.e., 12/03/2021.

meter and check meter. Billing process is based on the meter installed by the utility and is based on the billing statement issued by the utility.

The schematic of the solar power plant is presented in the figure below.



The standard estimated lifetime of the project activity is considered as 25 years. This may increase depending on the operation & maintenance of the plant.

The technical specification of the project is outlined below:

Sr	Description of Items		Key Technical Specification
1	Connected Capacity		35 MWAC
2	Solar PV Module	Manufacturer	SUNTECH/STP 330-24/Vfw_1500V
		PV module technology	Poly Crystalline Silicon
		Unit Nom. Power	330 Wp
		No of Module	In Series- 30 modules In parallel -4584 strings Total number of PV modules - 137520
		Number of PV Module	1,24,000± 10%
		Output of each PV Module	320,325,330 Watt
3	Grid tied Inverter	Manufacturer	SUNGROW/SG3125HV
		Inverter technology	Central inverters
		No of Inverter	12
		Operating Voltage	875-1300 V
		Frequency range	50Hz±5%
		Total Harmonic Distortion	<3%
4	Collector Plane Orientation		14^0
5	Energy Meter (Main Meter)		SL. No. LG242334949 Type/Make - Landis+Gyr E650 Number -1
6	Energy Meter (Check meter)		SL. No. LG242334947 Type/Make - Landis+Gyr E650 Number -1

Key Parameters

Annual production probability (P90)	61558 MWh

Corresponding Plant Load Factor	20.078%
Average Annual Degradation	0.7% per year
Average Life time of the Project activity	20 Years 00 Months

There is not technology transfer involved in the project activity.

Emission Reductions from anthropogenic sources:
The project activity is a greenfield intervention generating clean energy and supplying electricity generated to the Bangladesh Grid thereby displacing an equivalent amount of electricity which would have otherwise been generated by fossil fuel dominant grid connected power plant. The Project Proponent plans to avail the VCS benefits for the Project.

1.12 Project Location

The project activity is located in Paturia under Manikganj district in Bangladesh. Detailed location is presented below:

Upazila - Shivalaya

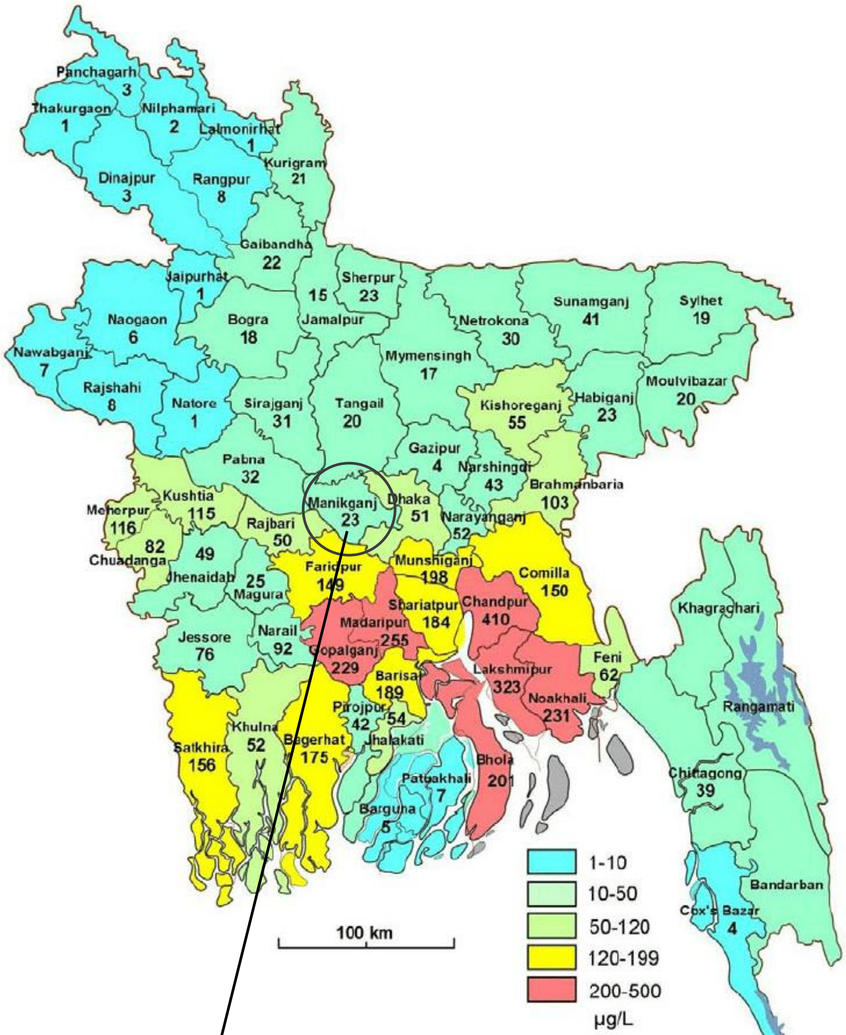
District – Manikganj

Country – Bangladesh

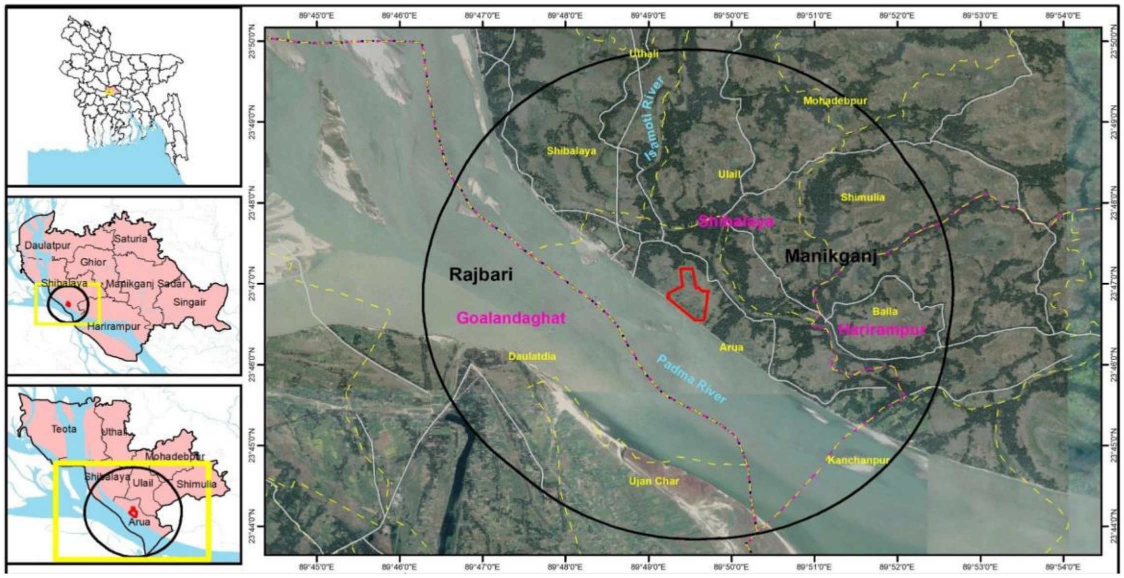
Geo-coordinate-

Latitude: 23°46'46.82"N ;

Longitude: 89°49'42.08"E



Bangladesh Map



1.13 Conditions Prior to Project Initiation

The project activity is a greenfield initiative involving deployment of 35 MW solar power projects to produce clean electricity by harnessing renewable energy (solar energy) resource. In the absence of the project activity, the equivalent amount of power would have generated by the operation of grid connected power plants and by the addition of new generation sources. Hence, the baseline for the project activity is the equivalent amount of power of the Bangladesh grid that are displaced due to the project activity. Since the baseline scenario is the same as the conditions existing prior to the project initiation therefore refer to Section 2.4 (Baseline Scenario) for detailed elaboration of baseline scenario.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The Project has received necessary approvals for development and commissioning of 35 MW Solar Power Project from the government of Bangladesh authorized nodal agencies including:

1. Licence by Bangladesh Electricity Regulatory Commission in favour of generation of 35 MW (AC,Solar PV) electricity by Spectra Solar Park Limited (Independent Power Producer (IPP)) as per sections 27 & 28 of the Bangladesh Energy Regulatory Commission Act, 2003 and regulation 16 of the Bangladesh Energy Regulatory Commission License Regulations, 2006.
2. Environmental Clearance by Department of Environment(DoE) Bangladesh.

Renewable Energy policy (Renewable Energy Policy of Bangladesh-2008) and other key acts and policies like Bangladesh Energy Regulatory Commission Act, 2003 and Bangladesh Electricity Act 2018 are framed primarily to encourage renewable energy power projects but does not influence the choice of fuel used for power generation. Moreover, there is no legal requirement on the choice of a particular technology or fuel for power generation. Implementation of Solar Power projects is therefore a voluntary activity and is not mandated by any legal mandates in Bangladesh.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project activity has not been registered, nor is it seeking registration under any other GHG program.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project is not included in an emissions trading program or any other mechanism that includes GHG allowance trading. The undertaking is submitted for same.

1.16.2 Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit. The undertaking is submitted for same.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

The project activity includes implementation 35 MW Grid connected Solar Power Project in Bangladesh. Electricity generated from the project activity is supplied to Bangladesh Power Development Board, thus substitutes high carbon intensive grid electricity by clean and renewable based electricity. Installation of the project activity will enhance the current renewable energy share of 1.04% in the national grid as well as contribute towards reducing greenhouse gas emission which would have other wise been emitted due to generation of electricity in the fossil fuel powered power projects. The project will also result in job creation both on temporary basis (during the construction stage) and permanent basis (during operation phase) including employability of youth.

SDG Goal	Initiatives	Expected contribution to sustainable development
SDG 7 – Ensure access to affordable, reliable,	The project activity will result in generation of electricity by harnessing solar energy which	The project will increase substantially the share of renewable energy in the energy mix

sustainable and modern energy for all	will in turn be fed to the national grid of Bangladesh.	of Bangladesh National Grid from renewable based electricity generated by the project activity and fed to the grid.
SDG 8 - Promote sustained, inclusive and sustainable economic growth, full and productive employment and descent work for all	Implementation (including construction) and operationalisation of the project activity required involvement and engagement of human resources.	The project activity has resulted in employability both on temporary basis (during construction) and permanent basis (during operation) including employment of youth.
SDG 13 - Take urgent action to combat climate change and its impacts	The electricity generated from the project activity that is fed to the grid will reduce generation of electricity in the fossil fuel-based grid connected power project.	Feeding of renewable based electricity to the grid will replace generation of electricity from fossil fuel-based power projects that is currently supplying electricity to the grid and reduce equivalent green house gas emission.

1.17.2 Sustainable Development Contributions Activity Monitoring

The Project activity includes implementation of 35 MW grid connected solar power project that feds in the electricity generated to the grid and thereby increase the share of renewable in existing national grid. Feeding of renewable based electricity to the grid will result in refraining of generation of electricity in the fossil fuel-based power plant that currently supplies power to the grid resulting in reduction of equivalent amount of greenhouse gas emission. The project activity will also result in both temporary and permanent employability.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
Sequential row number	*SDG Target number*	*Number and text of SDG indicator or, if no official SDG indicator is applicable, user-defined indicator*	*Indicate the project's contribution to the SDG Indicator (implemented activities to increase or decrease)*	*Brief description of the quantifiable impact of the project's activities related to the SDG indicator, during the monitoring period.*	*Brief description of the cumulative quantifiable impact of the project's activities related to the SDG indicator, over the project lifetime.*
1)	7.2	Renewable energy share in the total final energy consumption.	The project being a renewable based power generation unit feeding power to the grid will enhance the renewable energy share in the final energy consumption	The project has contributed to the SDG goals by supplying 54,128 MWh of renewable electricity to the grid	Likely supply of 1416 GWh of electricity to the national grid during the project lifetime.
2)	8.5	Average hourly earnings of female and male employees, by occupation, age and persons with disabilities	The project activity will result in temporary and primary employment opportunity during the construction and operation phase	The project has contributed to the SDG goals by creating employment opportunities for 25 persons	Likely creation of employment opportunity for 30 individuals

	8.6	Proportion of youth (aged 15-24 years) not in education, employment or training	The employment opportunities created through the project will result in employment of youths	The project has contributed to the SDG goals by creating employment opportunities for 5 youths	Likely creation of employment opportunity for 10 youths
3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to result in avoidance of greenhouse gas emission by refraining generation of electricity in fossil fuel-based power plant connected to the grid	Generation of 54,128 MWh of renewable electricity that is fed to the grid the project activity will result in avoidance of 35,643 tCO_{2e} during the monitoring period	Prevented an estimated release of 932 thousand tonnes of carbon-di-oxide into the atmosphere (in 25 years)

1.18 Additional Information Relevant to the Project

Leakage Management

Not Applicable to the project activity.

As per approved methodology ACM0002 (Version 20.0), no leakage is considered for the proposed project.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description

Further Information

No situation is identified that have a bearing on the eligibility of the project, the net GHG emission reductions or removals, or the quantification of the project's net GHG emission reductions or removals.

2 SAFEGUARDS

2.1 No Net Harm

No potential negative environmental or socio-economic impacts have been identified for the project and the impacts identified in the project are of minor significance. Project Proponent has conducted detailed Environmental and Social Impact Assessment (ESIA) to assess the Environment, Social and economic impacts of the solar PV based power project activity during its construction & operation phases to ensure the appropriate mitigation action be planned to be in place if there is any negative impact.

The ESIA clearly indicated that the environmental impact of the project activity is of minor significance. The ESIA outlines the social and economic benefits (as identified) linked to employment generation, local procurement, encouragement of local enterprise development and skill development within the communities from the project activity. The project in its entirety is likely to bring prosperity and development into the region and pave the way for greater decarbonisation of national grid.

In addition, the project activity is contributing towards reducing the GHG emission by replacing equivalent amount of electricity generated from fossil fuel fired power plants and contributing towards energy self-sufficiency.

2.2 Local Stakeholder Consultation

Local stakeholder consultation was carried out as a part of the ESIA study during 13th,14th and 15th February 2019.

Approach of Stakeholders consultation

1. Focused group discussion
2. Key Informant Interview

Stakeholders identified

1. land sellers,
2. subcontractors
3. local labourers
4. local community;
5. agricultural labourers
6. Union Chairman;
7. Government Department; and
8. NGO at Arua and Ulail Union.

Consultations was conducted during the ESIA with the objective of obtaining view of the project stakeholders about the planned project including their feedbacks, and concerns through individual interviews and focused group discussions (were undertaken at 10 locations). The ESIA ensured that, any grievance, suggestions or general feedback are accepted and addressed in a timely manner and incorporated in the project where applicable and relevant.

As a part of the consultation process, the project representatives explained all stakeholders about the type of development through the project and the technology used for power generation towards enhancing stakeholders' awareness about the project. The stakeholders were also made aware about the employment opportunities and other benefits likely to be rendered through this project and the importance of setting up solar PV power plant, not only to tap the cleaner sources for energy generation but most to meet the ever-increasing energy demand of state and its contribution towards fighting against climate change.

The project owner also shared a grievance register during individual interviews during focused group discussions where the stakeholder can put down his/her complain and the same if found genuine were to be addressed immediately. Also, regular stakeholder engagement is one the key focus at the site.

The project stakeholders were totally in support for setting up of these kinds of projects in the region.

The communities and the community representatives in and around study area were aware about and in favour of the proposed solar power project and expressed their assurance for support and cooperation for project activity. They didn't seem to have any objections or problem related to the development of solar power project in their area.

The community representatives and seller of land are of expectation in improvement of their livelihood from the upcoming project and aspired for development in the locality with the upcoming solar power project. The land sellers conveyed that land was taken from them for a greater cause, which they appreciate.

There were no comments raised by the stakeholders and they were totally in support for setting up of these kinds of projects in the region.

2.3 Environmental Impact

No negative environmental impacts have been identified from the project (based on the ESIA undertaken by the project proponent) and environmental impact assessment (EIA) is not required for the project. The project proponent has obtained environmental clearance for the project activity.

2.4 Public Comments

To be obtained.

2.5 AFOLU-Specific Safeguards

This section is not applicable as the project is a non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Methodology Title and reference: ACM0002 Grid-connected electricity generation from renewable sources --- Version 20.0[6]

The applied methodology draws upon the following tools:

Tool 07 - Tool to calculate the emission factor for an electricity system (Version 07.0, EB 100, Annex 4) [7]

Tool 32- Positive lists of technologies (Version 03.0, EB 110, Annex 4)[8]

3.2 Applicability of Methodology

The project activity is a green field grid interactive solar power project with cumulative capacity of 35 MW. Since the project is a 35 MW renewable energy project, therefore Large-scale Consolidated Methodology ACM0002: Grid-connected electricity generation from renewable

[6] https://cdm.unfccc.int/methodologies/DB/XP2LKUSA61DKUQC0PIWPGWDN8ED5PG
[7] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf
[8] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-32-v3.0.pdf

sources applies to project activities that include retrofitting, rehabilitation (or refurbishment), replacement or capacity addition of an existing power plant or construction and operation of a Greenfield power plant is applied. The applicability of the methodology is established through assessment of the following conditions.

Para No.	Applicability Conditions	Justification of eligibility
3	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units; (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	The project activity involves construction and operation of grid-connected solar power (renewable energy) project at a site where no renewable energy power plant was operated prior to the implementation of the project activity and therefore a “green field power project” as per the definition of the methodology and hence complies to the applicability condition (3.a). Hence the project activity meets the applicability condition of the methodology.
4	The methodology is applicable under the following conditions: (a) The project activity may include renewable energy power plant/unit of one of the following types: hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	The project activity involves construction and operation of greenfield grid-connected solar power project and hence complies to the applicability condition (4.a) of the methodology. Since the project activity does not include capacity additions, retrofits, rehabilitations or replacements of existing plant/unit, the applicability condition 4.b is not applicable/relevant for the project activity.

Para No.	Applicability Conditions	Justification of eligibility
5	In case of hydro power plants, one of the following conditions shall apply:[9] (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m2; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m2; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m2, all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m2; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the	The project activity involves construction and operation of greenfield grid-connected solar power project using solar energy for generation of electricity hence the applicability condition "5" is not applicable/relevant to the project activity as the applicability conditions is related to hydro power projects.

9 Project participants wishing to undertake a hydroelectric project activity that results in a new reservoir or an increase in the volume of an existing reservoir, in particular where reservoirs have no significant vegetative biomass in the catchments area, may request a revision to the approved consolidated methodology.

Para No.	Applicability Conditions	Justification of eligibility
	project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m2 shall be: (a) Lower than or equal to 15 MW; and (b) Less than 10 per cent of the total installed capacity of integrated hydro power project.	
6	In the case of integrated hydro power projects, project proponent shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal	The project activity involves construction and operation of greenfield grid-connected solar power project using solar energy for generation of electricity, hence the applicability condition “6” is not applicable/relevant to the project activity as the applicability conditions is related to hydro power projects.

Para No.	Applicability Conditions	Justification of eligibility
	flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
7	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	The project activity involves construction and operation of greenfield grid-connected solar power project using solar energy for generation of electricity hence the applicability condition 7" is not relevant as the same pertains to switching from fossil fuels to renewable energy sources or biomass fired power plants/units.
8	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	The project activity involves construction and operation of greenfield grid-connected solar power project using solar energy for generation of electricity hence the applicability condition 8 is not relevant as the same pertains to retrofits, rehabilitations, replacements, or capacity additions.
9	In addition, the applicability conditions included in the tools referred to below apply	The applicability of the tools is outlined below

Tool to calculate the emission factor for an electricity system	
Applicability Conditions	**Justification of eligibility**
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	This condition of applicable, OM, BM and CM are estimated using this tool under section 4.1 for calculating of the baseline emission.
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the	Since the project activity is grid connected, the condition is applicable and emission

Tool to calculate the emission factor for an electricity system	
Applicability Conditions	**Justification of eligibility**
project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	factor has been calculated accordingly.
In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in Bangladesh, a Non-Annex I country. Therefore, this condition is not applicable to the project activity.
Under this tool, the value applied to the CO2 emission factor of biofuels is zero.	The project activity is a green field solar power plant and hence the condition of biofuel emission factor is not applicable.

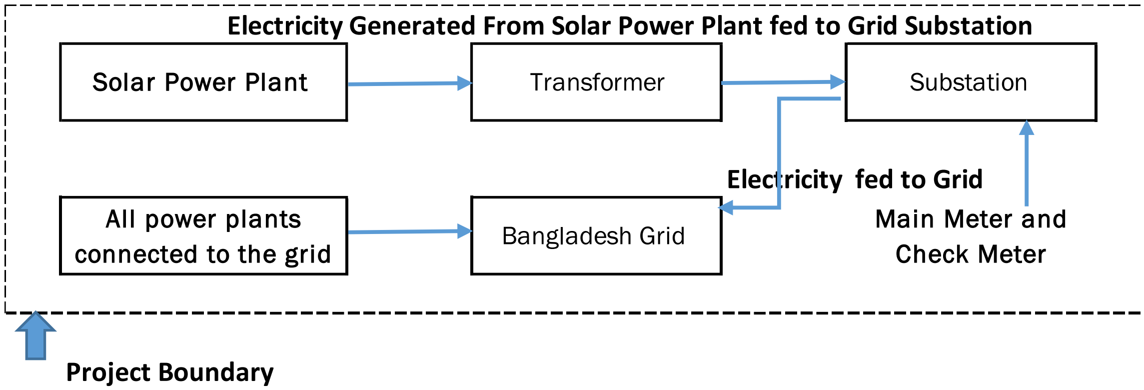
1. Compliance to applicability conditions of Methodological Tool "Tool for the demonstration and assessment of additionality - Version 7.0.0 is demonstrated in section 3.5 below

2. Compliance to applicability conditions of Methodological Tool Investment analysis - Version 11.0 is demonstrated in section 3.5 below

3. Compliance to applicability conditions of Methodological Tool common practice - Version 3.1 is demonstrated in section 3.5 below

3.3 Project Boundary

The spatial extent of the project boundary includes the project power plant / unit and all power plants / units connected physically to the electricity system that the project power plant is connected to. The project boundary therefore includes the project site where the power plant has been installed, associated power evacuation infrastructure, energy metering points, switch yards and other civil constructions and the connected Bangladesh Grid.



The project boundary includes CO_{2e} as greenhouse gas (GHG) that would have emanated from the operation of grid connected fossil fuel-based power plants in the absence of the project activity.

The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions.

Source		Gas	Included?	Justification/Explanation
Baseline	CO2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity	CO_2	Yes	Main emission source
		CH_4	No	Minor emission source
		N_2O	No	Minor emission source
Project	Solar Power Project	CO_2	No	As a zero-emission grid connected Solar power project no CO2 emissions will result from the project activity in operation phase.

3.4 Baseline Scenario

As per the approved consolidated Methodology ACM0002 (Version 20.0) para 22: “If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system (Version 07.0)”.

The project activity involves setting up of solar power projects to harness the renewable energy (solar energy) resource to produce electricity and supply it to the Bangladesh grid. In the absence of the project activity, the equivalent amount of power would have generated by the operation of existing grid connected power plants (fossil fuel dominated) and by the addition of new generation sources. Hence, the baseline for the project activity is the equivalent amount of power of the Bangladesh grid that are displaced due to the project activity.

The combined margin ($EF_{grid, CM,y}$) is the result of a weighted average of two emission factor pertaining to the electricity system: the operating margin (OM) and build margin (BM).

The combined margin of the Bangladesh grid used for the project activity is as follows:

Parameter	Value	Nomenclature	Source
$EF_{grid, CM,y}$	0.6585 tCO_{2e}/MWh	Combined margin CO_2 emission factor for the project electricity system in year y	Calculated as the weighted average of the operating margin (0.75) & build margin (0.25) values,
$EF_{grid, OM,y}$	0.6745 tCO_{2e}/MWh	Operating margin CO_2 emission factor for the project electricity system in year y	Calculated as the last 3-year weighted average (2017-18, 2018-19 and 2019-20) in accordance to Tool to calculate the emission factor for an electricity system (Version 07.0)" based on Data available from Bangladesh Power Development Board (Annual Report 2017-18,2018-19, 2019-20, and CDM Baseline Bangladesh Department of Environment (DoE) December 2012)
$EF_{grid,BM,y}$	0.6107 tCO_{2e}/MWh	Build margin CO_2 emission factor for the project electricity system in year y	Calculated in accordance to Tool to calculate the emission factor for an electricity system (Version 07.0)" based on Data available from Bangladesh Power Development Board (Annual Report 2019-20 and CDM Baseline Bangladesh Department of Environment (DoE) December 2012)

During the implementation of the project activity, the relevant national and/or sectoral policies, regulations and circumstances are taken into account.

- Implementation of solar PV based power generation unit for electricity generation is not mandatory under any law in Bangladesh, the project activity is thus a voluntary action.
- Despite the increase in renewable energy sources (including solar energy) in power sector, still about ninety-eight percent of installed power generation capacity is based on fossil-fuel based energy sources, hence the electricity grid is fed by electricity generated predominantly in fossil-fuel based power plants.

Construction of Solar PV based power plants are exempted from Environmental Impact Assessment (EIA). The Project has however received approval from Bangladesh Electricity Regulatory Commission and Department of Environment (DoE) Bangladesh.

3.5 Additionality

The project activity generates electricity using solar energy which is a renewable, zero emission source of energy. However, the project activity does not comply to the criteria of positive list as provided in CDM Tool 32: "Methodological Tool – Positive List of Technologies" and hence additionality of the project activity is demonstrated through a project specific additionality test.

Baseline considerations for the project are based on approved consolidated baseline methodology ACM0002 (Version 20.0). The methodology requires the project investor to determine the additionality based on "Methodological Tool- Tool for the demonstration and assessment of additionality", Version 7.0.0. The tool provides a step-wise approach to demonstrate and assess the additionality of a project (figure below). These steps are:

(a) Step 0 Demonstration whether the proposed project activity is the first-of-its-kind;

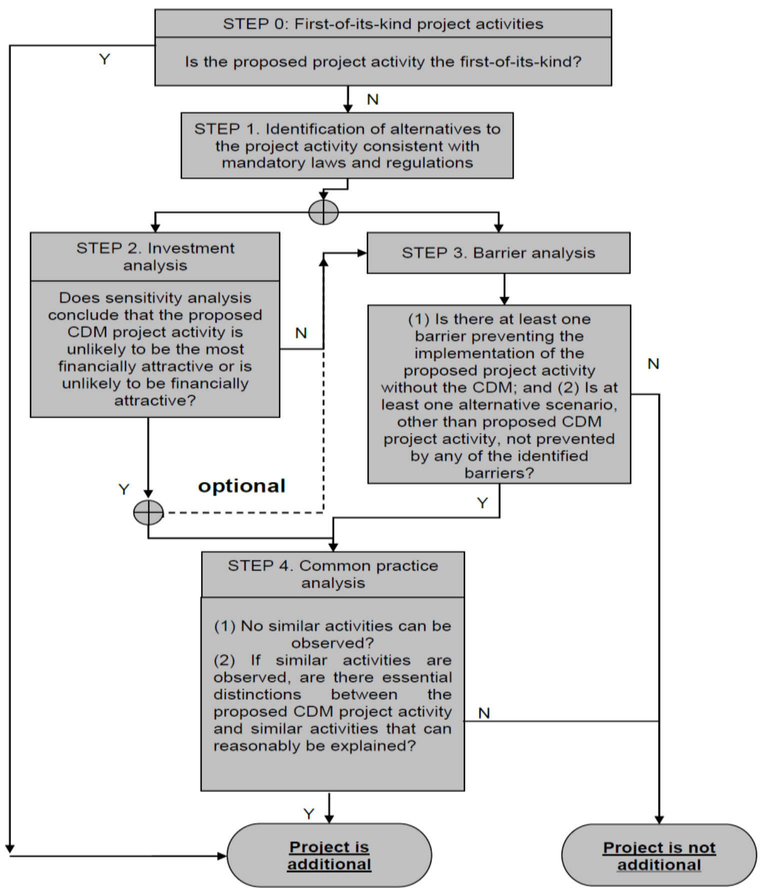
(b) Step 1 Identification of alternatives to the project activity;

(c) Step 2 Investment analysis;

(d) Step 3 Barriers analysis; and

(e) Step 4 Common practice analysis

Figure 1 Flowchart of the step-wise approach



The step-wise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs:

Step 0: Demonstration whether the proposed project activity is the first-of-its-kind

The project activity is a large-scale solar power project undertaken in Bangladesh. This is not the first such project to be installed in the country and therefore project activity does not meet this criterion.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity- Identify realistic and credible alternative(s) available to the project participants or similar project developers that provide outputs or services comparable with the proposed CDM project activity.

The proposed project activity is aimed at generating electrical energy using renewable energy (solar) resources and feed the electricity generated to the grid. Hence, the following alternatives are considered:

Alternative 1: The proposed project activity undertaken without being registered under VERRA.

The option of proposed project activity being undertaken without being registered under VERRA is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, this is not a plausible option as deployment of the project activity faces significant financial barriers established in the subsequent section. The electricity generated from the project activity would have been fed to the Bangladesh National Grid, which has a number of power plants including existing and upcoming ones which would be used in the absence of project activity. In the absence of project activity, power from the same grid will continue to provide for the requirement of the country.

Alternative 2: Continuation of the current situation (No proposed project activity and equivalent amount of energy would have been produced by the grid electricity system through its currently running power plants and by new capacity addition to the grid)

In such case the project owner would have continued without investment in project activity and continued with its usual business activities. In such case grid would continue with the fossil fuel-based power projects and this would result in GHG emissions. Hence, existing power plant and the new capacity add-on from a fossil fuel-based power plant is appropriate, realistic & credible baseline alternative for the project activity.

Outcome of Sub-step 1.a: Identified realistic and credible alternative scenario(s) to the project activity

Of the two alternatives outlined above, the first alternative is not possible as project activity is not viable without carbon credit benefits and second alternative is the baseline scenario for the project activity as per methodology. Therefore, continuation of the current situation is the likely alternative to the project activity.

The project being a green field project activity the baseline scenario in line with the methodology is “the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system Version 7.0”.

Sub-step 1b: The alternative(s) shall be in compliance with all mandatory applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g. to mitigate local air pollution.

Installation of a large-scale solar power project including the one proposed as project activity is consistent with mandatory laws and regulations Bangladesh and has obtained necessary clearance/approval.
Outcome of Sub-step 1.b: Identified realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations taking into account the enforcement in the region or country and EB decisions on national and/or sectoral policies and regulations

Hence, both the alternatives identified above are found to comply with the mandatory laws and regulations taking into account the enforcement of the legislations in the region/country and EB decisions on national and/or sectoral policies and regulations. However, Alternative 2 "Continuation of the current situation" has been selected as the appropriate baseline alternative for this project activity in line with methodology.

Step 2: Investment analysis

Investment analysis is carried out to determine on whether the proposed project activity is economically or financially less attractive than at least one other alternative, identified in step 1, without the revenue from the sale of emission reductions credits. This is demonstrated in in line with the Tools for sections as per "Investment Analysis" (Ver 11.0)

Sub-step 2a: Determine appropriate analysis method

The project activity envisages export of power to Bangladesh National grid and the revenues from the sale would be generated in accordance with the terms and tariffs established in the Power Purchase Agreement (PPA). Thus, simple cost analysis (Option I) cannot be used as the analysis method as the sale of the units of generated electricity shall result in a revenue stream during the operations of the Project activity. In the absence of the project activity continued use of grid electricity would have been the best plausible options and it does not require an investment. Hence investment comparison analysis (Option II) is also not appropriate for the project activity.

After eliminating Option, I and Option II, the use of Benchmark analysis (Option III) is the method of analysis that has been selected as the most suitable method. This method determines the attractiveness of the project activity for the investors, as well as provides a measure of the viability of the investment to generate revenues during its operation, as compared with other avenues and investment options. Hence, the Benchmark analysis method is to be employed for analysis of the said project.

Sub-step 2b (Option III): Apply benchmark analysis

The investment analysis using Benchmark analysis approach (Option III) has been chosen. Further, this method illustrates that the evaluation of the project by the project owner was carried out before the decision to undertake the project was taken and management approval had been granted.

Choice of Financial Indicator:

According to the "Tool for demonstration and assessment of Additionality" Version 07.0.0, the financial indicator can be based either on (1) project IRR or (2) equity IRR. There is no general preference between the approaches (1) or (2). The benchmark chosen for analysis shall be fully consistent with the choice of approach. Therefore, in accordance with the guidance, the relevant financial indicator for project activity has been chosen as post tax equity IRR.

Default Value Benchmark:

The cost of equity is determined by selecting the values provided in the Appendix, i.e. Default values for cost of equity (expected return on equity) is presented below:

Appendix of the CDM TOOL 27: Investment Analysis Version 11.0 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in Bangladesh = 11.72%

The Required return on equity (benchmark) was computed in the following manner:

Nominal Benchmark = {(1+Real Benchmark) *(1+Inflation rate)}-1

Where:

Default value for Real Benchmark = 11.72% (as per Appendix of CDM TOOL 27: Investment Analysis)

Inflation Rate forecast for Bangladesh.

Benchmark Estimation

Inflation Rate (at the time of investment decision making)

Inflation forecast as per ADB	Sources Link
7.10%	Projected average inflation

Benchmark

Description	Value	Sources Link
Default Value for Bangladesh as per UNFCCC guidelines	10.55%	https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v11.0.pdf
Inflation forecast (Projection from 2023)	7.10%	https://www.adb.org/countries/bangladesh/economy
Benchmark	19.65%	Calculated

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III): The Post tax Equity IRR is evaluated for the entire lifetime of the project activity, i.e. 25 years. It is calculated based on the cash outflows from and cash inflows into the project activity.

The IRR and Benchmark analysis is calculated in excel spreadsheet and is also summarized below. Based on result of IRR excel spreadsheets, equity IRR is assessed to be less than Benchmark. This substantiates that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark). Thus, it can be easily concluded that project activity is additional & is not business as usual scenario.

Investment Analysis

Details of the project		**Source**
Location	Manikganj	
Total Capacity (MW)	35	As per DPR
Expected Date of Commissioning	1-Mar-21	
Life of the plant (Yrs.)	20	As per DPR
Generation of electricity		
PLF (%)	20.08%	As per DPR
Annual generation (kWh)	6,15,58,000	Year 1
Annual Degradation from 2nd year	0.70%	As per DPR
Tariff rate at the decision making (USD/kWh)	0.139	As per PPA
Cost of Electrical Appliance	34,73,280	DPR (estimated @10% of project cost)
Period of Replacement	5 year	
Cost of Inverter	69,46,560	DPR (estimated @20% of project cost)
Period of Replacement	8 years	
Expenditure		
O & M Expenses	2,97,500	
O & M free for (Yr.)	-	
Escalation in the operational expenses (%)	8.00%	
Insurance	2,77,862	
Administrative Expenditure	1,72,800	
Financial parameters		
Total Cost (USD)	5,20,64,160	
Loan Amount (USD)	3,64,44,912	
Equity Investment (USD)	1,56,19,248	
Term loan		
Loan Amount (USD)	3,64,44,912	DPR
Interest rate (%)	6.50%	DPR
Loan Tenure (Qtr.)	40	DPR
Moratorium Period (Qtr.)	-	
Repayment Period (Qtr.)	40	DPR
Repayment instalments value (USD)	9,11,123	DPR
1st instalment from (Qtr. end)	1-Jun-20	DPR
Working Capital		
Receivables Period	2	Month

Details of the project		**Source**
Maintenance Spare @15% of the O&M expenses		
O&M Expenses	2	Month
Interest on Working Capital Debt	10.00%	
Depreciation		
Land		
Gross Depreciable Value (USD)	3,47,32,800	Calculated Value
Salvage Value (%)	10.00%	
Salvage value (USD)	34,73,280.00	Calculated Value
Net Depreciable Value (USD)	3,12,59,520	Calculated Value
Residual Value (USD)	34,73,280.00	Calculated Value
Depreciation rate	20%	https://assets.kpmg/content/dam/kpmg/xx/pdf/2018/12/bangladesh-2018.pdf
Tax rates		
Income tax rate (%)	0.00%	1st 15 year
Income tax rate (%)	37.50%	after 15 years / https://assets.kpmg/content/dam/kpmg/xx/pdf/2018/12/bangladesh-2018.pdf

Sub-step 2d: Sensitivity Analysis

Addressing Guidance under Sensitivity Analysis Version-11.0, following factors has been subjected to sensitivity analysis:

1. PLF
2. O&M Cost
3. Project Cost
4. Tariff

The rationale of sensitivity is, "The ultimate objective of the sensitivity analysis is to determine the likelihood of the occurrence of a scenario other than the scenario presented, in order to provide a cross-check on the suitability of the assumptions used in the development of the investment analysis."

Sensitivity Analysis

The results of sensitivity analysis show that even with a variation of +10% & -10% in project cost, O&M cost, PLF and Tariff Rate Equity IRR is significantly lower than the benchmark. And it is evident from the results given above; the project remains additional even under the most favourable conditions.

Probability to breach the benchmark:
Sensitivity Parameter 1 : PLF
PLF considered in financials for is as per Third Party report in line with “Guidelines for the reporting and validation of Plant load factors” stated in EB 48 Annex11 option 3(b). Hence, variation in PLF of more than 10% is unlikely to happen as the PLF has been reported as per the Third-Party Report based on long term data.

Sensitivity Parameter 2: O&M
The sensitivity analysis reveals that O&M will breach the benchmark at negative values and is hypothetical case. Since the O&M cost is generally subject to escalation and also subject to inflationary pressure, any reduction in the O&M costs is highly unlikely. Hence, the reduction in the O&M cost is highly unlikely.
Sensitivity Parameter 3 : Project Cost
Project Cost for financial analysis is considered based on DPR (Detailed Project Report) prepared at the time of investment decision making to go ahead with the project activity. Sensitivity is carried out for 10% variation and for threshold level below which benchmark is not breached. Thus, it is unlikely that project cost will change beyond sensitivity range
Sensitivity Parameter 4 : Tariff Rate
The tariff is fixed for entire lifetime of the project activity. Hence, there is no probability to get variation for the same. However, Sensitivity is carried out for +/-10% even then the benchmark is not breached.

Result of Sensitivity Analysis

Sensitivity Analysis	Equity IRR			
Variation %	**-10%**	**Normal**	**10%**	**Breaching Value**
PLF	7.83%	11.70%	15.67%	19.72%
O&M	11.94%	11.70%	11.46%	-360.09%
Project Cost	14.92%	11.70%	8.80%	-20.30%
Tariff Rate	7.83%	11.70%	15.67%	19.72%

Outcome of Step 2:

Final Result

Equity IRR without CDM	Benchmark (Equity IRR)
11.704%	19.65%

The above result establishes that the equity IRR is less than Benchmark.

This substantiates that, the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark). Thus, it can be easily concluded that, the project activity is additional & is not business as usual scenario.

Step 4: Common practice analysis

As per para 57 of Tool for demonstration and assessment of additionality" (Version 07.0.0), Step 2 analysis shall be complemented with an analysis of extent to which the proposed project type (e.g., technology or practice) has already diffused in the relevant sector and region. This test is a credibility check to complement the investment analysis.

Common Practice Analysis – 35 MW Solar Power Project

Step (1): Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.

Range	Capacity	Unit
+ 50%	52.5	MW
Capacity of the proposed project activity	35	MW
-50%	17.5	MW

Step (2): Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

- The projects are located in the applicable geographical area;
- The projects apply the same measure as the proposed project activity;
- The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;
- The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant;
- The capacity or output of the projects is within the applicable capacity or output range calculated in
- The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity, whichever is earlier for the proposed project activity.

Analysis of Step 2

Identification of the similar projects (CDM and non-CDM) is carried out as per sub-steps of Step (2) as follows:

- As the project is located Bangladesh therefore, projects in the geographical area of Bangladesh have been chosen for analysis.
- The project activity is a green-field solar power project and uses measure (b) “Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies”. Therefore, projects applying same measure (b) are candidates for similar projects.
- The energy source used by the project is solar. Hence, only solar energy projects have been considered for analysis.
- The project produces electricity; therefore, all power plants that produce electricity are candidates for similar projects.
- The capacity range of the projects is within the applicable capacity range from 17.5 MW to 52.5 MW.
- As per the methodological tool the projects that have started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity, are to be selected for consideration. The start date however resembles to start date definition of CDM project activity which is “the date on which the project participants commit to making expenditures for the construction or modification of the main equipment or facility (e.g. a wind turbine), or for the provision or modification of a service (e.g. distribution of energy-efficient light bulbs, change of transport management system), for the CDM project activity or CPA. Where a contract is signed for such expenditures, it is the date on which the contract is signed. In other cases, it is the date on which such

expenditures are incurred” In line with the start date definition of the CDM the cut off-date for investment analysis is considered as investment decision date.

Findings of analysis of Step 2

Numbers of Similar projects identified, which fulfil above-mentioned conditions have been identified form the annual report of Bangladesh Power Development Board. Base don’t he published information there is only one solar power plant within the capacity range.

1. Teknaf 20MW PP (Solartech)

Step (3): Within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing project verification. Note their number N_{all}.

Project activities, which have got registered or are under project verification with CDM/VCS/GS/GCC have been excluded in this step. After excluding the registered and under project validation projects the total number of projects.

N_{all} – 0

Step (4): Within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N diff.

As per the tool on Common Practice, the project activities have been separated from the different technologies on the basis two criteria:

Different technologies - are technologies that deliver the same output and differ by at least one of the following (as appropriate in the context of the measure applied in the proposed clean development mechanism (CDM) project activity and applicable geographical area):

(a) Energy source/fuel (example: energy generation by different energy sources such as wind and hydro and different types of fuels such as biomass and natural gas);

(b) Feed stock (example: production of fuel ethanol from different feed stocks such as sugar cane and starch, production of cement with varying percentage of alternative fuels or less carbon-intensive fuels);

(c) Size of installation (power capacity)/energy savings:
(i) Micro (as defined in paragraph 24 of decision 2/CMP.5 and paragraph 39 of decision 3/CMP.6);
(ii) Small (as defined in paragraph 28 of decision 1/CMP.2);
(iii) Large.

(d) Investment climate on the date of the investment decision, inter alia:
(i) Access to technology;
(ii) Subsidies or other financial flows;
(iii) Promotional policies;
(iv) Legal regulations.

(e) Other features, inter alia:
(i) Nature of the investment (example: unit cost of capacity or output is considered different if the costs differ by at least 20 %).

The Project is deployed under preferential tariff regime (Investment climate on the date of the investment decision, inter alia- Promotional policies) so solar power project that are not covered under preferential tariff/promotional policy are considered as different from the project activity. However, there is only 1 project so project with different measures is considered as zero.

N-$_{diff}$ – 0

Step (5): Calculate factor F= 1-Ndiff/Nall representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.
Calculate
F= 1 – N diff /N all
F= = 1
So, F is greater than 0.2

Nall – Ndiff = 1
So Nall – Ndiff is less than 3

As per the Tool for common practice analysis, version 03.1, the proposed project activity is a common practice within a sector in the applicable geographical area if both the following conditions are fulfilled:
(a) the factor F is greater than 0.2, and
(b) Nall-Ndiff is less than 3.

Since the value of factor Nall – Ndiff for the proposed project activity is 1 which is less than 3, the project activity is not a “common practice” within sector in the applicable geographical area.

3.6 Methodology Deviations

The project did not apply any methodology deviations

4 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

As per approved consolidated methodology ACM0002, version 20.0, emission reduction is estimated as difference between the baseline emission and project emission after factoring into leakage.

$ER_y = BE_y - PE_y$

As the project activity is a solar project, there won't be any leakage emissions from the project activity. Hence, $LE_y = 0$

Where:

ER_y	Emission reductions in year y (t CO2e/yr)
BE_y	Baseline emissions in year y (t CO2/yr)
PE_y	Project Emission in year y (t CO2e/yr)
LE_y	Leakage Emission in year y (t CO2e/yr)

As per the approved consolidated Methodology ACM0002, version 20.0, Baseline Emissions include only CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid- connected power plants. The baseline emissions are to be calculated as follows:

$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$

Where:

BE_y	Baseline emissions in year y (t CO2/yr)
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	Combined margin CO2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system Version 7.0" (t CO2/MWh)

As per ACM0002, V 20.0, If the project activity is the installation of a Greenfield power plant, then:

$EG_{PJ,y} = EG_{facility,y}$

$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EG_{facility,y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

The net electricity exported to Bangladesh Power Development Board (BPDB) from the project is measured & monitored with the help of energy meter (main meter) installed within the project boundary. There is also a backup meter to measure the export & import of electricity with BPDB. Monthly Joint Meter reading

As per methodology, combined grid emission factor is estimated as per the "Tool to calculate the emission factor for an electricity system" version 07 is presented below:

As per Methodological tool: Tool to calculate the emission factor for an electricity system (Version 07.0, EB 100, Annex 4), following six steps have been followed:
(a) Step 1: Identify the relevant electricity systems;
(b) Step 2: Choose whether to include off-grid power plants in the project electricity system (optional);
(c) Step 3: Select a method to determine the operating margin (OM);
(d) Step 4: Calculate the operating margin emission factor according to the selected method;
(e) Step 5: Calculate the build margin (BM) emission factor;
(f) Step 6: Calculate the combined margin (CM) emission factor.

Step 1: Identify the relevant electricity systems

As described in tool "For determining the electricity emission factors, identify the relevant project electricity system. Similarly, identify any connected electricity systems". Project participants may delineate the project electricity system using any of the following options:

Option A: "A delineation of the project electricity system and connected electricity systems published by the DNA or the group of the DNAs of the host country(ies), In case a delineation is provided by a group of DNAs, the same delineation should be used by all the project participants applying the tool in these countries;

Option 2: A delineation of the project electricity system defined by the dispatch area of the dispatch centre responsible for scheduling and dispatching electricity generated by the project activity. Where the dispatch area is controlled by more than one dispatch centre, i.e. layered dispatch area, the higher level area shall be used as a delineation of the project electricity system (e.g. where regional dispatch centres are required to comply with dispatch orders of the national dispatch centre then area controlled by the national dispatch centre shall be used);

Option 3. A delineation of the project electricity system defined by more than one independent dispatch areas, e.g. multi-national power pools.

Keeping the options into consideration, Department of Environment (DoE), Ministry of the Environment and Forest (MoEF) of Bangladesh, has considered the national grid of Bangladesh while developing national baseline on power and energy sector[10]. **Power Grid Company of Bangladesh** is the sole grid operating agency and wheels the entire grid connected power. It is a government owned company operating in Bangladesh and a subsidiary of Bangladesh Power Development Board.

Step 2: Choose whether to include off-grid power plants in the project electricity system

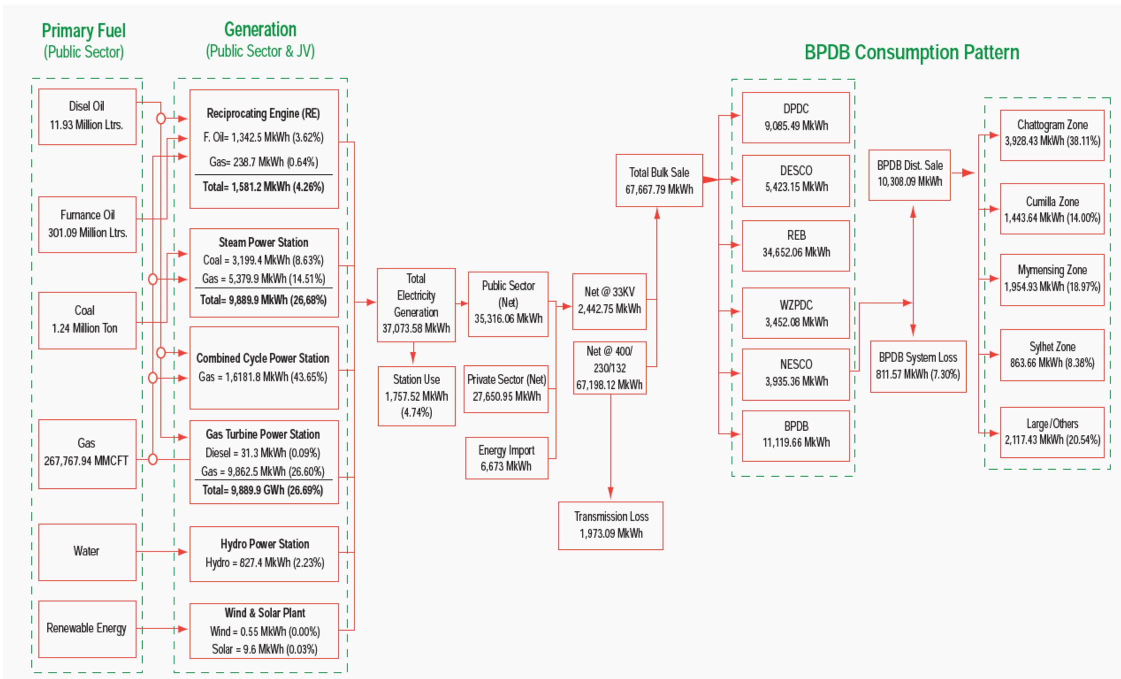
[10]Clean Development Mechanism(CDM) baseline, Department of Environment, Ministry of Environment

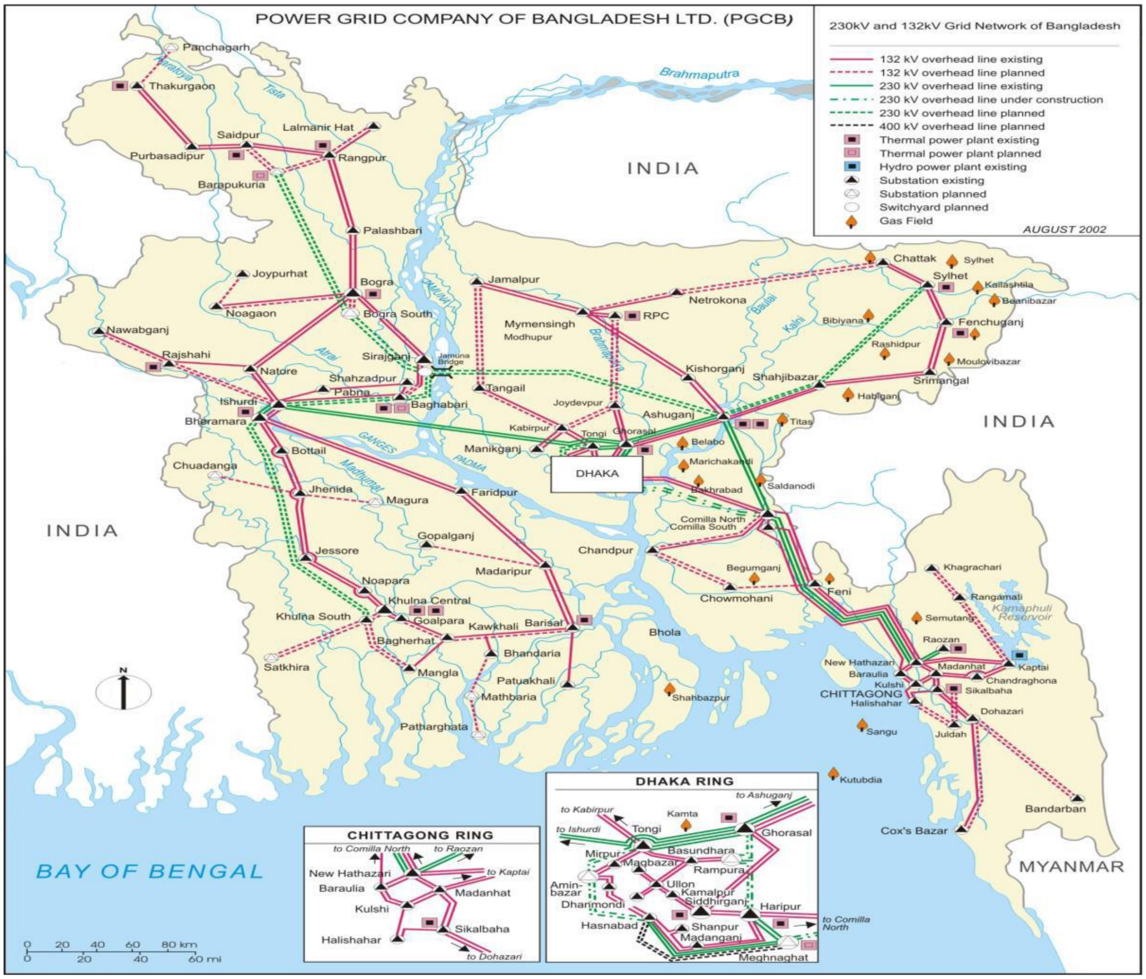
Project participants may choose between the following two options to calculate the operating margin and build margin emission factor:

Option I- Only grid power plants are included in the calculation.

Option II- Both grid power plants and off-grid power plants are included in the calculation

The Project Participant has chosen only grid power plants in the calculation.





Step 3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor (EFgrid,OM,y) is based on one of the following methods, which are described under Step 4:

(a) Simple OM; or

(b) Simple adjusted OM; or

(c) Dispatch data analysis OM; or

(d) Average OM.

The data required to calculate Simple adjusted OM and Dispatch data analysis OM is not possible due to lack of availability of data to project developers. The choice of other two options for calculating operating margin emission factor depends on generation of electricity from low-cost/ must-run sources. In the context of the methodology low cost/must run resources typically include hydro, geothermal, wind, low-cost biomass, nuclear and solar generation.

In accordance to the tool the simple OM method can only be used if any one of the following requirements is satisfied:

1. Low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in: 1) average of the five most recent years, and the average of the five most recent years shall be determined by using one of the

approaches described under the tools; or 2) based on long-term averages for hydroelectricity production (minimum time frame of 15 years)

2. The average amount of load (MW) supplied by low-cost/must-run resources in a grid in the most recent three year

Share of Must-Run (Hydro/Nuclear) (% of Net Generation)

	2015-16	2016-17	2017-18	2018-19	2019-20
Total Generation (GWh)	52,193	57,275.97	62,677.91	70,533	71,418.87
Total Hydro (GWh)	962.20	982.04	1024.31	724.6487	825.19
Other Renewable (GWh)	0.13	0.09	3.79	38.6197	61.06
% Share of Net Generation	1.84%	1.71%	1.64%	1.08%	1.24%

Annual Report of Bangladesh Power Development Board

The above data clearly shows that the percentage of total grid generation by low-cost/ must-run plants (on the basis of average of five most recent years) for the Bangladesh grid is less than 50 % of the total generation. Thus, the Average OM method cannot be applied, as low cost/must run resources constitute less than 50% of total grid generation. The simple OM emission factor is calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (tCO_2/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units.

For the simple OM, the simple adjusted OM and the average OM, the emissions factor can be calculated using either of the two following data vintages:

(a) **Ex-ante option**: if the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required.

OR

(b) **Ex-post option**: if the ex-post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring.

Project proponent has chosen ex-ante option for calculation of Simple OM emission factor using a 3-year generation-weighted average, based on the most recent data available at the time of submission of the PD to the DOE for project validation. OM determined at project validation stage will be the same throughout the crediting period. There will be no requirement to monitor & recalculate the emission factor during the crediting period.

Step 4: Calculate the operating margin emission factor according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (t CO_2/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units.

The simple OM may be calculated by one of the following two options:

(a) **Option A:** Based on the net electricity generation and a CO_2 emission factor of each power unit; or

(b) **Option B**: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system. Option B can only be used if: (i) The necessary data for Option A is not available; and (ii) Only nuclear and renewable power generation are considered as lowcost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and (iii) Off-grid power plants are not included in the calculation (i.e. if Option I has been chosen in Step 2).

Option A: Calculation based on average efficiency and electricity generation of each plant

Under this option, the simple OM emission factor is calculated based on the net electricity generation of each power unit and an emission factor for each power unit, as follows:

$$EFgrid, OMsimple, y = \frac{\sum m\ EGm, y\ \times EFEL, m, y}{\sum m\ EGm, y}$$

Where:

$EF_{grid,OMsimple,y}$	Simple operating margin CO2 emission factor in year y (t CO2/MWh)
$EG_{m,y}$	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	CO2 emission factor of power unit m in year y (t CO2/MWh)
m	All power units serving the grid in year y except low-cost/must-run power units
y	The relevant year as per the data vintage

Determination of $EF_{EL,m,y}$

Option A1 - If for a power unit m data on fuel consumption and electricity generation is available, the emission factor (EFEL,m,y) should be determined as follows:

$$EFEL, m, y = \frac{\sum i\ FCi, m, y \times NCVi, y \times EFCO2, i, y}{EGm, y}$$

Where:

$EF_{EL,m,y}$	CO2 emission factor of power unit m in year y (t CO2/MWh)
$FC_{i,m,y}$	Amount of fuel type i consumed by power unit m in year y (Mass or volume unit)
$NCV_{i,y}$	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$	CO2 emission factor of fuel type i in year y (t CO2/GJ)

$EG_{m,y}$	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
m	All power units serving the grid in year y except low-cost/must-run power u nits
i	All fuel types combusted in power unit m in year y
y	The relevant year as per the data vintage

Option A2 - If for a power unit m only data on electricity generation and the fuel types used is available, the emission factor should be determined based on the CO2 emission factor of the fuel type used and the efficiency of the power unit, as follows:

$$EFEL,m,y = \frac{EFCO2,m,i,y \times 3.6}{\eta m,y}$$

Where:

$EF_{EL,m,y}$	CO_2 emission factor of power unit m in year y (t CO_2/MWh)
$EF_{CO2,m,i,y}$	Average CO2 emission factor of fuel type i used in power unit m in year y (t CO_2/GJ)
$\eta_{m,y}$	Average net energy conversion efficiency of power unit m in year y (ratio)
m	All power units serving the grid in year y except low-cost/must-run power units
y	The relevant year as per the data vintage
3.6	Conversion factor (GJ/MWh)

The Simple OM calculation and the data used for the calculation is provided in separate excel sheet. The resulting values of this calculation are as follows:

The operating margin emission factor has been calculated using a 3-year data vintage:

Net Generation in Operating Margin (GWH) (incl. Imports)			
	2017-18	2018-19	2019-20
Bangladesh Grid	62,686	70,567	71,435
Simple Operating Margin (tCO2/MWh) (incl. Imports) (1) (2)			
	2017-18	2018-19	2019-20
Bangladesh Grid	0.6956	0.6827	0.6480

Weighted Generation Operating Margin	
Bangladesh Grid	0.6745

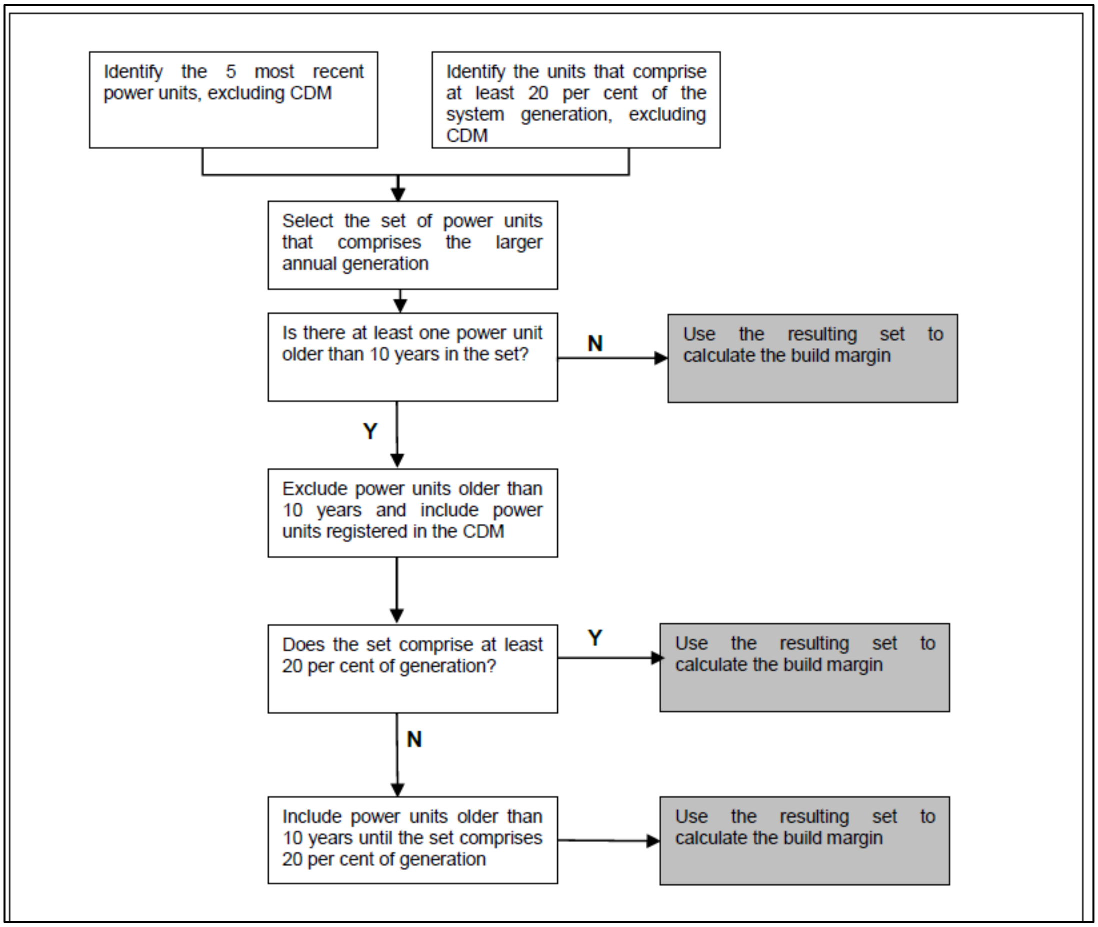
Step 5: Calculate the build margin (BM) emission factor

As per Methodological tool: “Tool to calculate the emission factor for an electricity system” (Version 07.0, EB 100, Annex 4) para 72: In terms of vintage of data, project participants can choose between one of the following two options:

(a) **Option 1** - for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group at the time of PDD submission to the DOE for project validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.

(b) **Option 2** - For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

Option 1 as described above is chosen by PP to calculate the build margin emission factor for the project activity. BM is calculated ex-ante based on the most recent information available at the time of submission of PD and is fixed for the entire crediting period.



The build margin emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which electricity generation data is available, calculated as follows:

$$EFgrid,BM,y = \frac{\Sigma EGm,y \times EFEL,m,y}{\Sigma EGm,y}$$

Where:

$EF_{grid,BM,y}$	Build margin CO2 emission factor in year y (t CO2/MWh)
$EG_{m,y}$	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	CO2 emission factor of power unit m in year y (t CO2/MWh)
m	Power units included in the build margin
y	Most recent historical year for which electricity generation data is available

Build Margin (tCO2/MWh) (not adjusted for imports

Build Margin (tCO2/MWh) (not adjusted for imports)	
	2019-20
Bangladesh Grid	0.6107

Step 6: Calculate the combined margin (CM) emission factor ($EF_{grid,CM,y}$)

As per Methodological tool: "Tool to calculate the emission factor for an electricity system" (Version 07.0, EB 100, Annex 4) para 81:

The calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

(a) Weighted average CM; or

(b) Simplified CM.

PP has chosen option (a) i.e weighted average CM to calculate the combined margin emission factor for the project activity.

The combined margin emissions factor is calculated as follows:

$EF_{grid,CM,y} = EF_{grid,OM,y} \times W_{OM} + EF_{grid,BM,y} \times W_{BM}$

Where:

$EF_{grid,BM,y}$	Build margin CO2 emission factor in year y (t CO2/MWh)
$EF_{grid,OM,y}$	Operating margin CO2 emission factor in year y (t CO2/MWh)
W_{OM}	Weighting of operating margin emissions factor (per cent)
W_{BM}	Weighting of build margin emissions factor (per cent)

The following default values should be used for WOM and WBM:

For solar project activities: WOM = 0.75 and WBM = 0.25 (owing to their intermittent and non-dispatchable nature) for the second crediting period and for subsequent crediting periods. Since project activity is of power generation by using solar, the above weightage has been considered for OM and BM.

Combined Margin Emission Factor for Wind/Solar Projects
Therefore, $EF_{grid,CM,y}$ = 0.6745 * 0.75 + 0.6107 * 0.25 = 0.6585 tCO_2/MWh

Baseline emission factor (EFy):

The baseline emission factor is calculated using the combined margin approach as described in Step 6 above:

Therefore, EFy = EFgrid,CM,y = 0.6585 tCO2/MWh

4.2 Project Emissions

The project emission calculation as per para 31of ACM0002 version 20

For most renewable energy power generation project activities, PEy = 0. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation:

$PE_y = PE_{FF} + PE_{GP} + PE_{HP,y}$

Where:

PE_y	Project emissions in year y (t CO2e/yr)
PE_{FF}	Project emissions from fossil fuel consumption in year y (t CO2e/yr)
PE_{GP}	Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y (t CO2e/yr)
$PE_{HP,y}$	Project emissions from water reservoirs of hydro power plants in year y (t CO2e/yr)

The project activity is a solar power project and in accordance to ACM0002 version 20, Para 33, “For all renewable energy power generation project activities, emissions due to the use of fossil fuels for the backup generator can be neglected” therefore

PEy = 0

4.3 Leakage

As per para 53 of ACM0002 version 20, No leakage emissions need to considered for the project activity.

4.4 Estimated Net GHG Emission Reductions and Removals

Formula used to calculate the net emission reduction for the project activity is:

ERy = BEy - PEy

As per para 53 of the applied methodology ACM0002 version 20.0, no other leakage emissions are considered. In this project activity.

Where:

ER_y	Emission reductions in year y (t CO2e/yr)
BE_y	Baseline emissions in year y (t CO2/yr)
PE_y	Project Emission in year y (t CO2e/yr)

Baseline Emission (BEy)

The baseline emissions is the product of electrical energy baseline $EG_{PJ,y}$ expressed in MWh of electricity produced by the renewable generating unit multiplied by an emission factor.

BEy = $EG_{PJ,y}$ * $EF_{grid,CM,y}$

Where:

BEy	Baseline emissions in year y (t CO2/yr)
EG PJ,,y	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh).

$EF_{grid,CM,y}$	Combined margin CO2 emission factor for grid connected power generation in year y calculated using the latest version of "TOOL07: Tool to calculate the emission factor for an electricity system" (t CO2/MWh)

For Green field project

$EG_{PJ,y} = EG_{facility,y}$

Where:

EG PJ,,y	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh).
$EG_{facility,y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Project Emission

As per para 31 of the applied methodology ACM0002 (version 20.0), no project emissions are considered in this project activity.

Hence, project emissions PEy = 0 tCO2e.

Leakage

As per methodology ACM0002 v20, para 53, no other leakage emissions are considered. Leakage emission as per methodology is "The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected".

Emission Reduction

Therefore, ER y = BE y

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
Year 1	40,535	-	-	40,535
Year 2	40,252	-	-	40,252
Year 3	39,970	-	-	39,970
Year 4	39,690	-	-	39,690

Year 5	39,412	-	-	39,412
Year 6	39,136	-	-	39,136
Year 7	38,862	-	-	38,862
Total	277,857	-	-	277,857

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid, OM, y}$
Data unit	tCO_2/MWh
Description	Operating Margin CO2 emission factor in the year y
Source of data	Annual Reports of Bangladesh Power Development Board www.bpdb.gov.bd/site/page/c4161d54-5b85-4917-a8d2-68a2d1b26dd4/মাসিক-বার্ষিক-প্রতিবেদন
Value applied:	0.6745
Justification of choice of data or description of measurement methods and procedures applied	Calculated as per "Tool to calculate the emission factor for an electricity system, version 07.0.0" as 3-year generation weighted average using data for the years 2017-18, 2018-19 & 2019-20. The data are obtained from "Annual Reports of Bangladesh Power Development Board.
Purpose of Data	Calculation of baseline emissions
Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	$EF_{grid, BM, y}$
Data unit	tCO_2/MWh
Description	Build Margin CO2 emission factor in the year y

Source of data	Annual Reports of Bangladesh Power Development Board www.bpdb.gov.bd/site/page/c4161d54-5b85-4917-a8d2-68a2d1b26dd4/মাসিক-বার্ষিক-প্রতিবেদন
Value applied:	0.6107
Justification of choice of data or description of measurement methods and procedures applied	Calculated as per “Tool to calculate the emission factor for an electricity system, version 07.0.0”. The data are obtained from “Annual Reports of Bangladesh Power Development Board.
Purpose of Data	Calculation of baseline emissions
Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	$EF_{grid, CM, y}$
Data unit	tCO_2/MWh
Description	Combined Margin CO2 emission factor in the year y
Source of data	Annual Reports of Bangladesh Power Development Board www.bpdb.gov.bd/site/page/c4161d54-5b85-4917-a8d2-68a2d1b26dd4/মাসিক-বার্ষিক-প্রতিবেদন
Value applied:	0.6585
Justification of choice of data or description of measurement methods and procedures applied	The combined margin emissions factor is calculated as follows: EFgrid,CM,y= EFgrid,OM,y* WOM + EFgrid, BM,y* WBM Where: EFgrid,BM,y= Build margin CO2 emission factor in year y (tCO2/MWh) EFgrid,OM,y= Operating margin CO2 emission factor in year y (tCO2/MWh) WOM = Weighting of operating margin emissions factor (%) = 75% WBM= Weighting of build margin emissions factor (%) = 25%
Purpose of Data	Calculation of baseline emissions
Comments	This parameter is fixed ex-ante for the entire crediting period.

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh /year
Description	Quantity of net electricity generation supplied by the solar project plant to the grid in year y
Source of data	Billing Statement - Meter Reading Report Jointly signed by BPDB official and PP representative.
Description of measurement methods and procedures applied	The Net electricity supplied to the grid by the project activity will be calculated based on the energy delivered to (exported) & received (imported) from the grid monitored and measured by Main Meter and backup Meter reading reports provided by BPDB. Cross Checking: Quantity of net electricity supplied to the grid will be cross checked from the Invoices
Frequency of monitoring/recording	Continuous measurement and monthly recording.
Value applied:	*60,280* MWh/year (*average annual generation for 3 years)*
Monitoring equipment	Energy Meter Main Meter SL. No. LG242334949 (Type Landis+Gyr E650) Back Meter SL. No. LG242334947 (Type Landis+Gyr E650)
QA/QC procedures applied	Calibration of all the meters will be undertaken at required intervals and faulty meters will be duly replaced immediately. The calibration frequency will be once in five years.
Purpose of data	Calculation of Baseline emissions
Calculation method	Not Applicable
Comments	Data will be archived in paper & electronically for a period of 2 years beyond the end of crediting period or of the last issuance of credits for this project activity, whichever occurs later.

5.3 Monitoring Plan

The monitoring plan is developed in accordance with requirement for VCS project activity and methodology. The monitoring plan is proposed for grid-connected solar power project being implemented in Bangladesh. The monitoring plan, is implemented by the project participant describes about the monitoring organisation, parameters to be monitored, monitoring practices, quality assurance, quality control procedures, data storage and archiving.

The authority and responsibility for monitoring, measurement, reporting and reviewing of the data rests with the project participant. Project participant proposed the following structure for data monitoring, collection and data archiving for this project activity. The team comprises of the following members:

VCS Project Monitoring Team ↓	1. Reviewing monthly and annual generation 2. Estimating GHG emission reduction 3. Identify opportunities for Future improvement 4. Addressing grievance related to the project activity
O&M Head ↓	1. Implementation of corrective measures of discrepancies identified in the reported parameters 2. Review of monthly report 3. Ensuring calibration of the monitoring equipment
Shift In charge ↓ Site Engineer	1. Monitoring and reporting of GHG performance related parameters following the guidelines provided in the project document 2.Preperation of monthly and annual generation report 3. Reporting of any discrepancies identified in the reporting parameters 4. Ensuring calibration of the monitoring equipment

Data Measurement

The export and import energy will be measured continuously using Main and Check meters located at the substations. Readings of meters shall be taken on monthly basis by authorized officer of BPDB in the presence of the representative of Project participant. Based on the Meter Reading Statement, invoices will be raised. These invoices can be used for cross checking the meter readings taken for the respective project activity.

Data Collection and Achieving

Readings from meters will be collected in the presence of the plant in-charge. Export and Import data would be recorded and stored in logs as well as in electronic form. The records are checked periodically by the Plant Manager and discussed thoroughly with the plant supervisor. The period of storage of the monitored data will be 2 years after the end of crediting period or till the last issuance of CERs for the project activity whichever occurs later.

Emergency Preparedness

The project activity will not result in any unidentified activity that can result in substantial emissions from the project activity. No need for emergency preparedness in data monitoring is visualized. In the event that the main meter, which is used to record the net electricity exported by the project, is found to be faulty it will be repaired or replaced by by authorized officer of BPDB and the data from the check meter will be used in its place. In the unlikely event that the check meter fails it will also be repaired or replaced

Personnel Training

In order to ensure a proper functioning of the project activity and a properly monitoring of emission reductions, the staff will be trained. The plant staffs will be trained in equipment operation, data recording, reports writing, operation and maintenance and emergency procedures in compliance with the monitoring plan.

Metering Arrangement

Line diagram with metering arrangement for the solar project activity can be identified form the plant SLD.

QA/QC Procedures

The energy meters at the feeders are maintained and owned by BPDB. The accuracy class of meters and calibration frequency is under purview of BPDB and PP do not have any control on it. The meters are calibrated by BPDB as per calibration requirement.

6 ACHIEVED GHG EMISSION REDUCTIONS AND REMOVALS

6.1 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh
Description	*Quantity of net electricity generation supplied by the solar project plant to the grid in year y*
Value applied:	*54127*
Comments	The relevant data information has been archived

6.2 Baseline Emissions

The baseline emissions is the product of electrical energy baseline EG PJ,y expressed in MWh of electricity produced by the renewable generating unit multiplied by an emission factor.

BEy = EG PJ,,y * EF grid,CM,y

Where:

BEy	Baseline emissions in year y (t CO2/yr)
EG PJ,y	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh).
$EF_{grid,CM,y}$	Combined margin CO2 emission factor for grid connected power generation in year y calculated using the latest version of "TOOL07: Tool to calculate the emission factor for an electricity system" (t CO2/MWh)

Here

EG PJ,y = 54127.746 MWh

EF $_{grid,CM,y}$ = 0.6585 tCO2/MWh

BEy = 54127.746 × 0.6585

= 35,643 tCO2

6.3 Project Emissions

The project activity involves in harnessing Solar power. So, the emissions from the project are zero

6.4 Leakage

The project activity involves in harnessing Solar power. So, the emissions from the project are zero

6.5 Net GHG Emission Reductions and Removals

As per the applied methodology, emission reductions are calculated as follows:

ERy = BEy – PEy

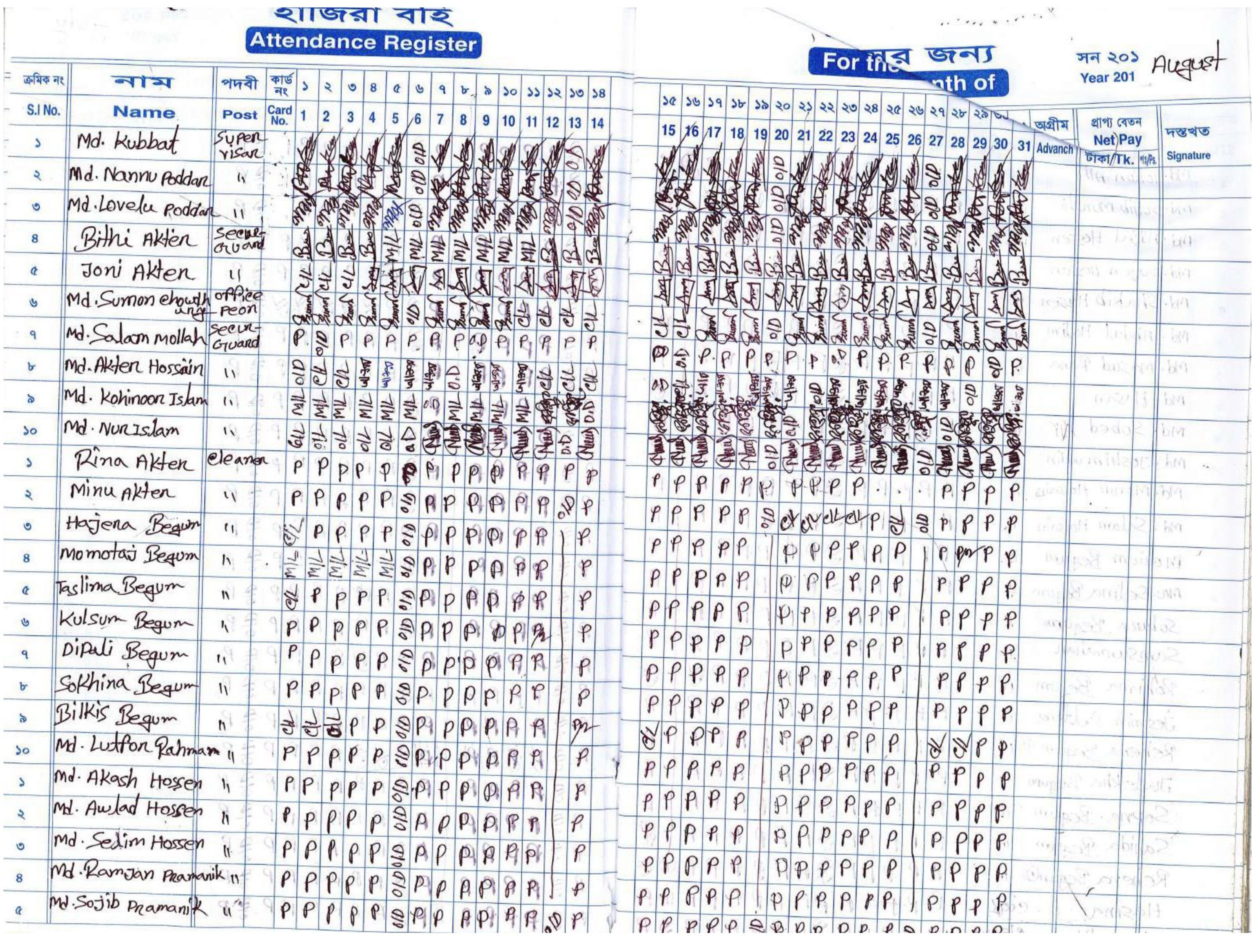
Where

ER_y	Emission reductions in year y (t CO2e/yr)
BE_y	Baseline emissions in year y (t CO2/yr)
PE_y	Project Emission in year y (t CO2e/yr)

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
12-03-2021 to 31-12-2021	32,508	0	0	32,508
1-1-2022 to 31-1-2022	3,135	0	0	3,135
Total	35,643	0	0	35,643

APPENDIX 1:

Copy of attendance register to demonstrate employment from the project activity



ক্রমিক নং S.I No.	নাম Name	পদবী Post
১	Md. Kubbat	Supervisor
২	Md. Nannu Poddar	〃
৩	Md. Lovelu Poddar	〃
৪	Bithi Akten	Secure Guard
৫	Joni Akten	〃
৬	Md. Sumon chowdhury	office Peon
৭	Md. Salam mollah	Secure Guard
৮	Md. Akten Hossain	〃
৯	Md. Kohinoor Islam	〃
১০	Md. Nur Islam	〃
১	Rina Akten	cleaner
২	Minu Akten	〃
৩	Hajena Begum	〃
৪	Momotaj Begum	〃
৫	Taslima Begum	〃
৬	Kulsum Begum	〃
৭	Dipali Begum	〃
৮	Sokhina Begum	〃
৯	Bilkis Begum	〃
১০	Md. Lutfor Rahman	〃
১	Md. Akash Hossen	〃
২	Md. Awlad Hossen	〃
৩	Md. Selim Hossen	〃
৪	Md. Ramjan Pramanik	〃
৫	Md. Sojib Pramanik	〃